

FictiPay Datathon

Round 2 — Advanced Challenge

Customer Lifetime & Survival Analysis

Context & Eligibility

Round 1 predicted whether a customer would churn within a fixed 30-day window. Round 2 asks a harder question: not just if — but when. You will model the full survival curve and instantaneous hazard rate per customer, enabling dynamic, time-sensitive retention scoring.

Dataset

Table	Key Columns	Notes
Transaction	TrxID, TRX_DATETIME, SRC_ACCOUNT, DST_ACCOUNT, TRX_TYPE, TRX_AMT	200M+ rows · January–March 2024 only
KYC	ACCOUNT_ID, ACCOUNT_TYPE, ACCOUNT_OPEN_DATE, GENDER, REGION	~2M rows · open dates may pre-date 2024
DayEndBalance	ACCOUNT_ID, DATE, AVAILABLE_BALANCE	~180M rows · January–March 2024

Churn definition: a customer churns on the date that marks 30 consecutive days without any transaction. Customers still active at June 30, 2024 are right-censored. All survival times T are measured from March 31, 2024 (observation cutoff). Account open dates in KYC may pre-date January 2024 — use them to compute tenure as a model covariate.

Understanding train_labels.csv

The training label file has three columns: ACCOUNT_ID, DURATION_DAYS, EVENT_FLAG. These are derived from the full January–June transaction history using the 30-day inactivity rule. Participants only receive January–March data and must predict these values for test customers.

Column	Values	Meaning
EVENT_FLAG	0 or 1	1 = customer churned (had 30 consecutive silent days before June 30) 0 = still active on June 30 — right-censored, true churn not observed
DURATION_DAYS	0 – 91	Days from March 31 until the churn event (EVENT=1) or until June 30 (EVENT=0) $T = (\text{last_txn_date} + 30 \text{ days} - \text{March } 31) \cdot \text{clip}(0, 91)$

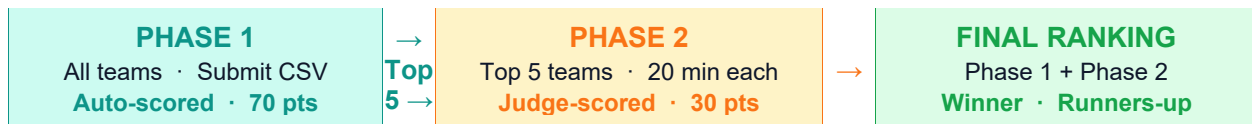
Worked example:

ACCOUNT_ID	DURATION_DAYS	EVENT_FLAG	What happened
CUST000001	14	1	Last txn March 17 → churn April 16 → T = 16 days. Churned early.
CUST000005	88	1	Last txn May 28 → churn June 27 → T = 88 days. Active deep into the window, churned just before study end.
CUST000009	91	0	Still active June 30. Censored at T = 91. True churn date unknown.
CUST000012	0	1	No transactions in Jan–Jun. Already churned before March 31.

DURATION_DAYS = 91 does not mean the customer transacted on June 30 — it means they were still active enough that no 30-day silence was observed before study end.
DURATION_DAYS = 0 means the customer had already been silent for 30+ days by March 31 and their churn predates the prediction window.

Competition Structure

All teams submit a predictions file. Phase 1 scores are computed automatically. The top 5 teams by Phase 1 score present their findings and business recommendations. The final ranking combines both phases.



PHASE 1 — Modelling Submission — Auto-Scored

70 pts

What to Build

Build a survival model that predicts when each customer will churn from their January–March transaction history. You are free to use any technique. The only thing scored is the quality of your test set predictions.

Submit predictions for all customers in test.csv.

Submission Format

Submit a single CSV with exactly five columns, one row per customer in test.csv:

```
ACCOUNT_ID,RISK_SCORE,SURV_PROB_30D,SURV_PROB_60D,SURV_PROB_90D
CUST00000000000001,0.823,0.41,0.28,0.19
CUST00000000000002,0.041,0.96,0.91,0.87
```

Column	Range	Description
ACCOUNT_ID	—	Customer ID from test.csv
RISK_SCORE	$[0, \infty)$	Higher = higher churn risk. Used for c-index. Can be predicted hazard, log-hazard, or any monotone risk score.
SURV_PROB_30D	$[0, 1]$	P(survive past day 30 from March 31) = P(active on April 30).
SURV_PROB_60D	$[0, 1]$	P(survive past day 60 from March 31) = P(active on May 30).
SURV_PROB_90D	$[0, 1]$	P(survive past day 90 from March 31) = P(active on June 29).

Survival probabilities must be monotone decreasing: $\text{SURV_PROB_30D} \geq \text{SURV_PROB_60D} \geq \text{SURV_PROB_90D}$ for every row.

Automatic Scoring Formula

Two metrics are combined into a single Phase 1 score. Both are computed against the private solution the moment you submit.

Metric 1 — Concordance Index (c-index) 40 pts

$\text{Score} = \max(0, (\text{c_index} - 0.5) / 0.5) \times 40$

0.5 = random ranking · 1.0 = perfect ranking · uses RISK_SCORE column

Metric 2 — Integrated Brier Score (IBS) 30 pts

$\text{Score} = \max(0, (0.05 - \text{IBS}) / 0.05) \times 30$

0.05 = null model baseline at ~5% event rate · 0.0 = perfect calibration · uses SURV_PROB_* columns

$\text{Phase 1 Total} = \text{c-index score} + \text{IBS score}$

c-index	IBS	Score
0.50	0.050	0 pts
0.60	0.040	14 pts
0.70	0.025	31 pts
0.80	0.015	45 pts
0.90	0.005	59 pts

Tiebreaker: equal total scores are resolved by the higher c-index.

PHASE 2 — Final Presentation — Top 5 Teams

30 pts

The five teams with the highest Phase 1 scores present their work to the judging panel. Each team has 20 minutes: 15 minutes to present, 5 minutes for Q&A. Judges have 30 pts.

Final position : Phase1 score + Phase 2 score.(Or Ranked Choice on Phase 2)

What to Present

- Your modelling approach — which model you built, why you chose it, and what the survival curves show.
- Key hazard ratios or feature importances interpreted in plain language.
- Business Recommendations.